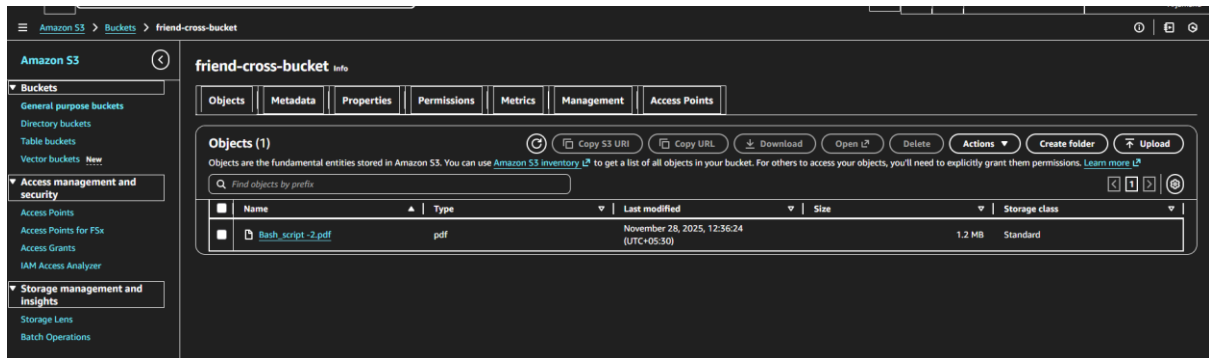


S3_Bucket

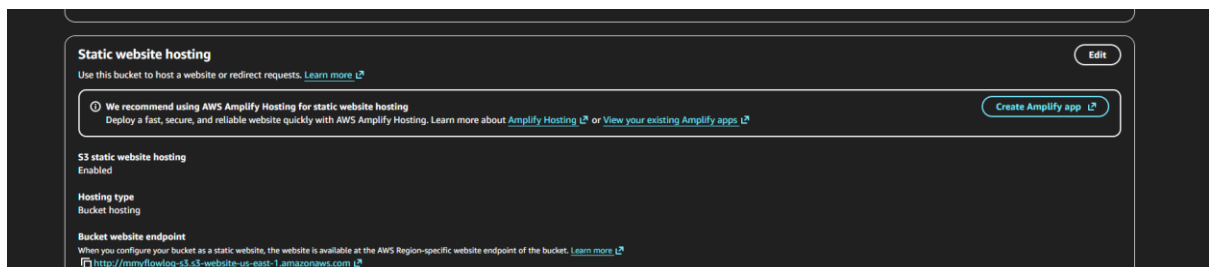
1. Create an S3 bucket and upload some objects to S3.

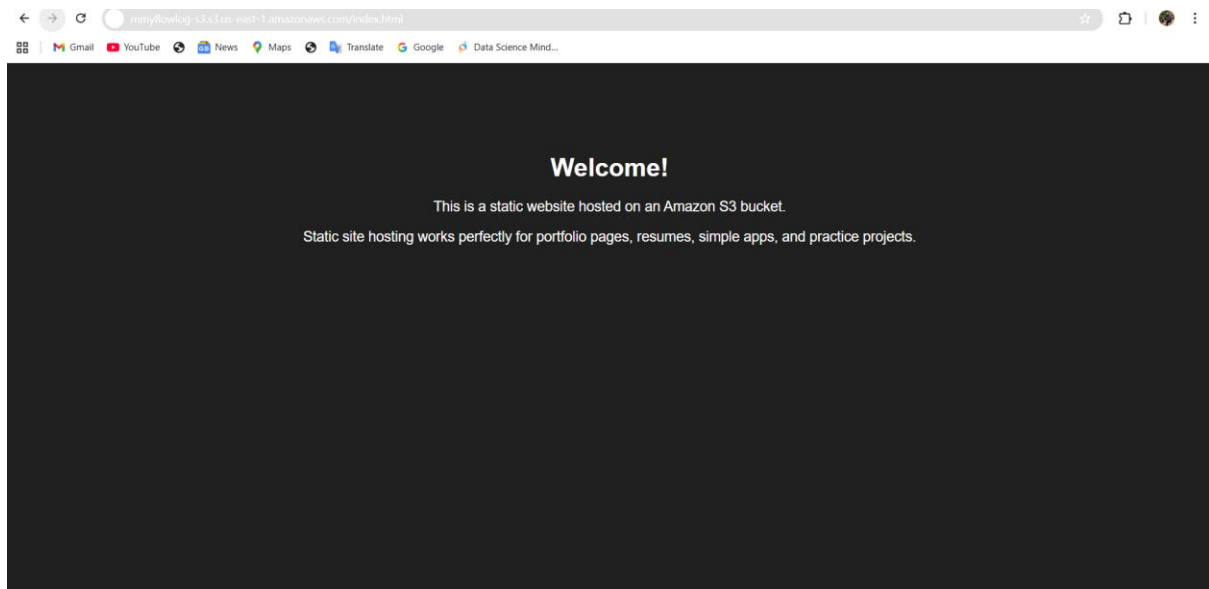
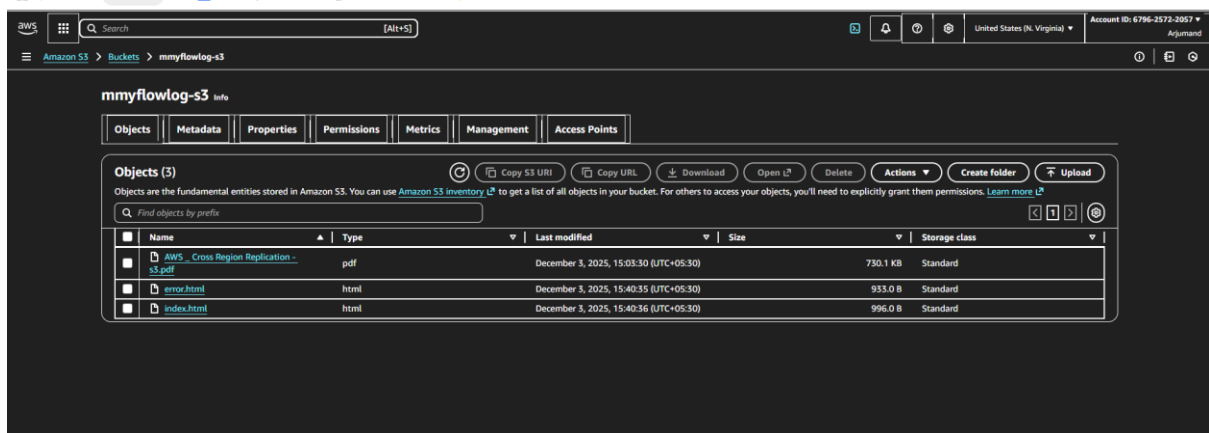
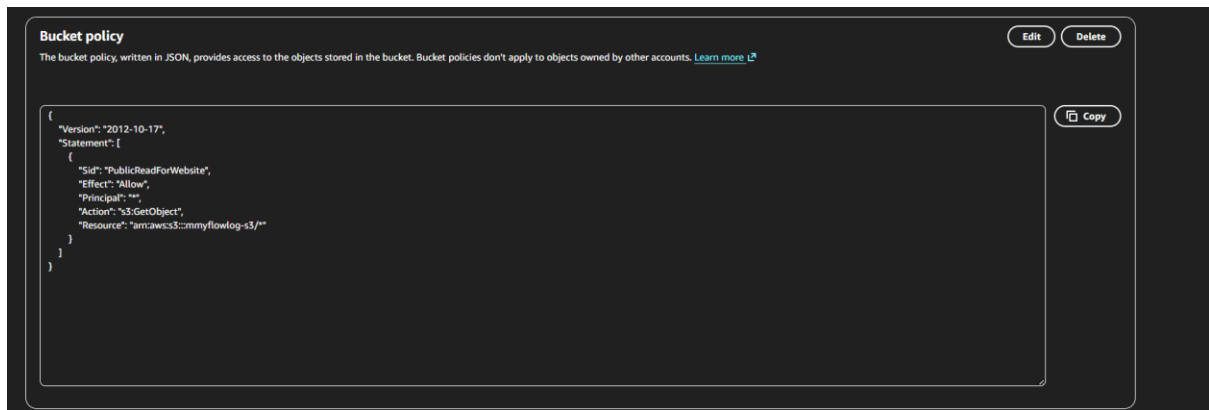
- I have created 1 bucket where I uploaded a file in it
- Go s3 > create bucket > select name > rest all will be default and create bucket
- Upload file > add file > select file > open file



2. Deploy a static website in the S3 bucket.

- I created an S3 bucket and enabled **Static Website Hosting** in the bucket settings.
- Then I prepared two files — **index.html** for the homepage and **error.html** for error handling.
- I opened the bucket in the AWS S3 console and went to the **Objects** section.
- Using the **Upload** button, I uploaded both HTML files directly into the bucket.
- Once uploaded, I made the files public and accessed the website using the S3 static website URL.





3. Enable cross-region replication on S3 buckets.

- First, I created two S3 buckets in different AWS regions — one as the source and one as the destination.
- Then I enabled **Versioning** on both buckets because CRR requires versioning to track changes.

- I opened the source bucket and went to **Management** → **Replication** → **Create rule**.
- I selected **Replicate all objects** and chose the destination bucket in another region.
- AWS asked for permissions, so I allowed it to create a new **IAM replication role** automatically.
- I kept the option **not to replicate existing objects**, so only new uploads will replicate.
- After saving the rule, any new object uploaded to the source bucket automatically gets copied to the destination bucket, confirming CRR is enabled.

Replication rules Info
Replication enables automatic and asynchronous copying of objects across buckets in the same or different AWS Regions. A replication configuration is a set of rules that define what options should be applied to a group of objects during replication.

Replication configuration settings Create replication job Edit
Configuration settings affect all replication rules in the bucket.

Source bucket
my-s3-bucket-ohio123

Source Region
US East (Ohio) us-east-2

IAM role
[s3crr_role_for_my-s3-bucket-ohio123](#)

Replication rules (1) View details Edit rule Delete Actions Create replication rule
Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. [Learn more](#)

	Replication rule name	Status	Destination bucket	Destination Region	Priority	Scope	Storage class	Replica owner	Replication Time Control	KMS-encrypted objects (SSE-KMS or DSSE-KMS)	Replica modification sync
●	CRR-Replication-Rule	Enabled	s3://my-bucket-friend	US East (N. Virginia) us-east-1	0	Entire bucket	Same as source	Same as source	Disabled	Do not replicate	Disabled

my-s3-bucket-ohio123 Info

Objects **Metadata** **Properties** **Permissions** **Metrics** **Management** **Access Points**

Objects (1) Copy S3 URL Copy URL Download Open Delete Actions Create folder Upload
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix Show versions

	Name	Type	Last modified	Size	Storage class
■	Configuring a static website on Amazon S3.pdf	pdf	December 3, 2025, 16:20:02 (UTC+05:30)	305.6 KB	Standard

my-bucket-friend Info

Objects **Metadata** **Properties** **Permissions** **Metrics** **Management** **Access Points**

Objects (1) Copy S3 URL Copy URL Download Open Delete Actions Create folder Upload
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Find objects by prefix Show versions

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■	Configuring a static website on Amazon S3.pdf	pdf	December 3, 2025, 16:20:02 (UTC+05:30)	305.6 KB	Standard

4. Configure a bucket policy so only the Admin user can see the objects of the S3 bucket.

- The Admin user was blocked because the bucket policy was denying all users due to a wrong condition key.
- The policy used **aws:PrincipalArn**, which does not work correctly for IAM users logging in through the console.
- Because of this, S3 treated the Admin user as unauthorized and denied the **s3:ListBucket** permission.
- To fix it, the condition was updated to **aws:username: "Admin"**, allowing only the Admin user to bypass the deny rule.
- The Admin IAM user also needed **AmazonS3FullAccess** permission to perform S3 actions.
- After updating both the bucket policy and IAM permissions, the Admin could list and access objects.
- Now the bucket correctly restricts access to only the Admin user while denying everyone else.

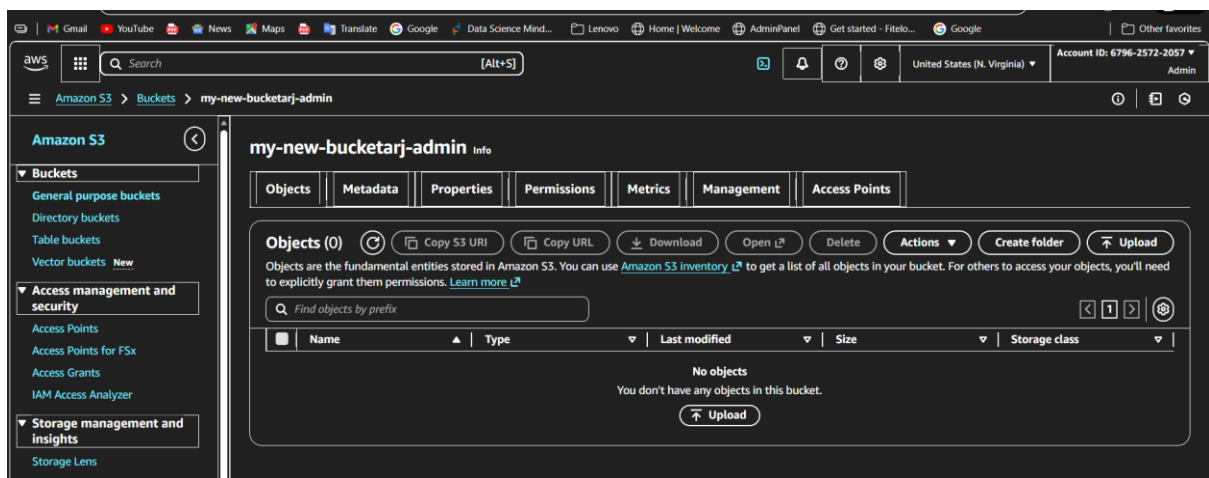
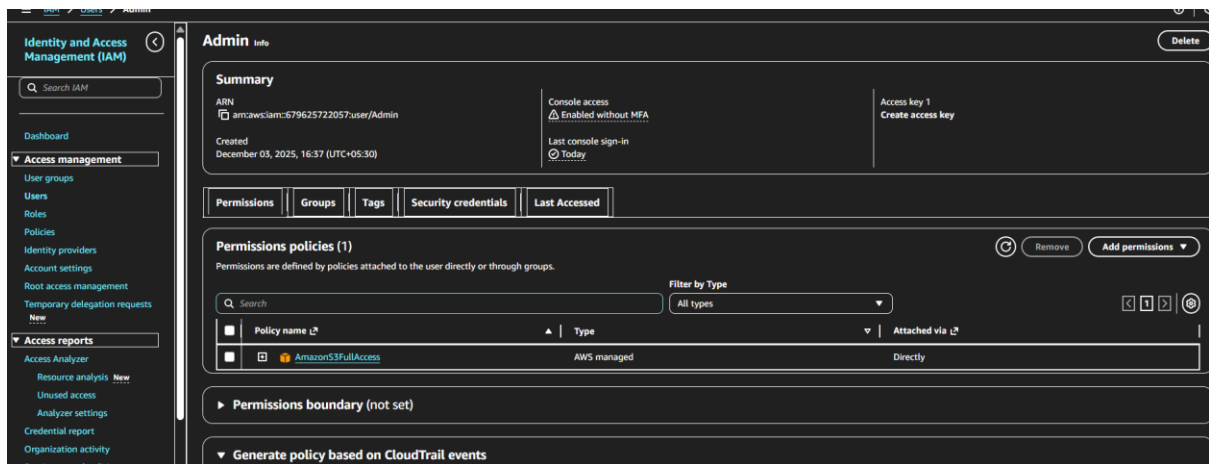
The screenshot displays the AWS Management Console interface for Amazon S3. The top section shows the 'my-new-bucket-aj-admin' bucket policy editor, which has been successfully updated. The policy JSON is as follows:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowDevopsUserOnly",
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": [
        "arn:aws:s3::my-new-bucket-aj-admin",
        "arn:aws:s3::my-new-bucket-aj-admin/*"
      ],
      "Condition": {
        "StringNotEquals": {
          "aws:PrincipalArn": "arn:aws:iam::679625722057:user/Admin"
        }
      }
    }
  ]
}
```

Below the policy editor, the 'Object Ownership' section is visible. The bottom section of the screenshot shows the 'Vector buckets' list, which includes the following buckets:

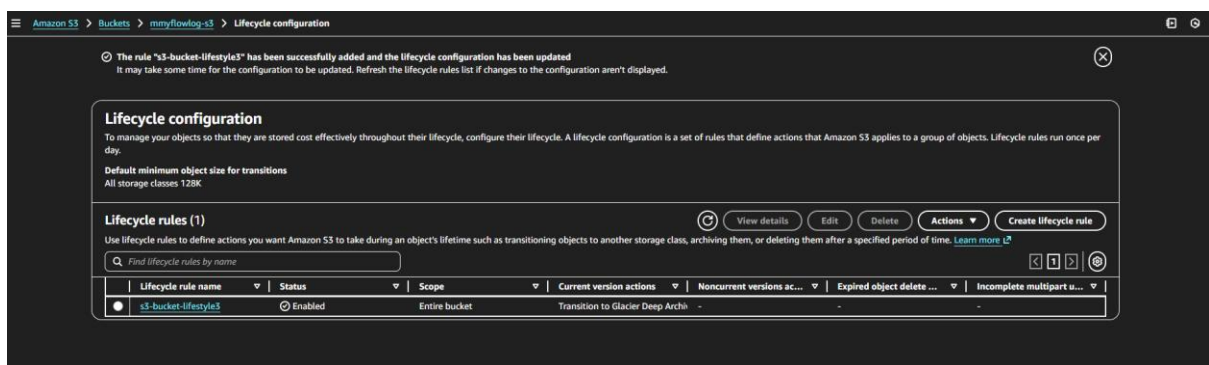
Name	Creation date
friend-cross-bucket	November 27, 2025, 19:29:59 (UTC+05:30)
mmyflowlog-s3	November 23, 2025, 00:25:46 (UTC+05:30)
my-bucket-friend	November 28, 2025, 12:46:25 (UTC+05:30)
my-new-bucket-aj-admin	December 3, 2025, 16:40:32 (UTC+05:30)
my-s3-bucket-ohio123	December 3, 2025, 15:55:40 (UTC+05:30)

On the right side of the bucket list, there is a 'Storage Lens' section providing visibility into storage usage and activity trends, including an 'External access summary - new info'.



5. Set up lifecycle policies to automatically transition or delete objects based on specific criteria.

- I opened my S3 bucket, went to the Management tab, and clicked Create lifecycle rule.
- I applied the rule to all objects and selected actions to transition objects to cheaper storage classes after specific days.
- I configured transitions—for example, moving objects to Standard-IA after 30 days and to Glacier after 90 days.
- Finally, I added an expiration rule to delete objects automatically after 365 days and saved the lifecycle policy.



6. Push some objects to S3 using the AWS CLI.

- Connected AWS with cli using access key and ID.
- Created one small file using vi editor
- Now moved that file to s3 bucket using command `aws s3 mv <filename> s3://<bucketname>/`

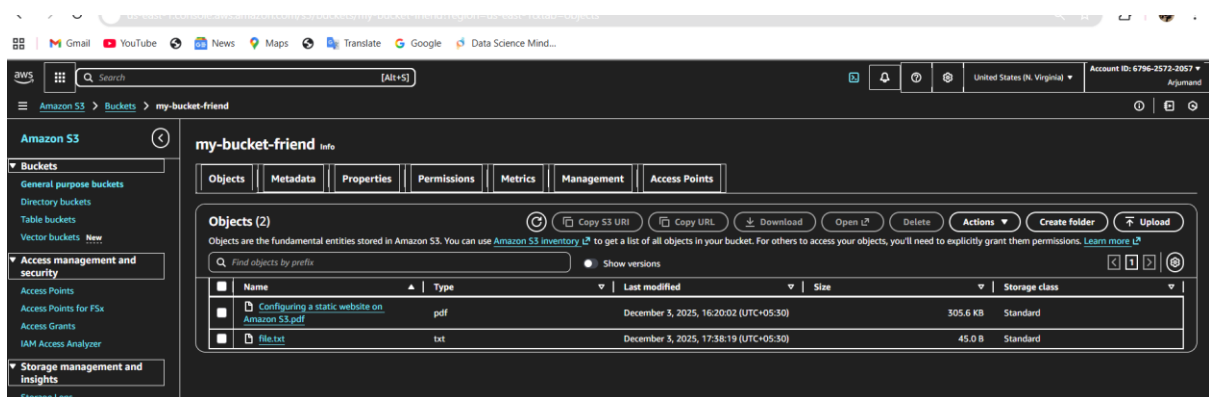
```
Arjunand@Arjunand MINGW64 ~/Downloads (master)
$ aws configure
AWS Access Key ID [*****U3AX]:
AWS Secret Access Key [*****Gxp]:
Default region name [us-east-1]:
Default output format [json]:

Arjunand@Arjunand MINGW64 ~/Downloads (master)
$ aws s3 ls
2025-11-27 19:29:59 friend-cross-bucket
2025-11-23 00:25:46 myFlowlog-s1
2025-11-28 12:46:25 my-bucket-friend
2025-12-03 16:40:32 my-new-bucketar-admin
2025-12-03 15:53:40 my-s3-bucket-ohio123

Arjunand@Arjunand MINGW64 ~/Downloads (master)
$ vi file.txt

ipradsSetup (6).exe *                               rustdesk-1.4.1-x86_64.exe*
ipradsSetup.exe*                                     script/
Oracle_VirtualBox_Extension_Pack-7.2.4.vbox-extpack

Arjunand@Arjunand MINGW64 ~/Downloads (master)
$ aws s3 mv file.txt s3://my-bucket-friend/
move: .\file.txt to s3://my-bucket-friend/file.txt
```



7. Write a Bash script to create an S3 bucket.

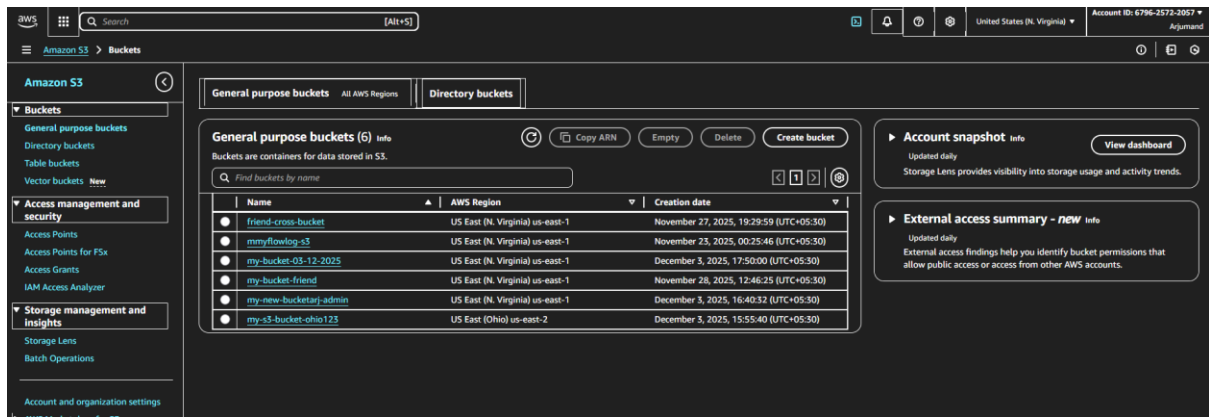
- I have connected with aws account with cli using access id and access key
- Created a bash script to create bucket
- Given execution permission to bash script
- I have run the bash script now it has created bucket called my-bucket-03-12-2025

```
#!/bin/bash

BUCKET_NAME="my-bucket-03-12-2025"
REGION="us-east-1"

aws s3api create-bucket \
  --bucket $BUCKET_NAME \
  --region $REGION

if [ $? -eq 0 ]; then
  echo "Bucket '$BUCKET_NAME' created successfully in region $REGION."
else
  echo "Bucket creation failed."
fi
```



8. Upload a 1 GB file to S3 using the CLI.

- I have uploaded 2.9GB file into my aws s3 bucket
- I have connected with aws account then moved folder from system downloads to s3 bucket using command 'aws s3 cp <folder_name> s3://bucketname+folder_name/ --recursive
- I went to s3 bucket and checked the status of upload

